

Write SQL query to solve the problem given below

Here we are talking about the Bank related information of a person.

For which you need to create three tables named as Bank, Account holder and Loan table. And solve the problem stated below.

Create a Bank table, attributes are : branch id, branch name, branch city Create a Loan table, attributes are : loan no, branch id, account holder's id, loan amount and loan type Create a table named as Account holder for the same scenario containing the attributes are account holder's id, account no, account holder's name, city,contact, date of account created, account status (active or terminated), account type and balance.

- Consider an example where there's an account holder table where we are doing an intra bank transfer i.e. a person holding account A is trying to transfer \$100 to account B. - for this you have to make a transaction in sql which can transfer fund from account A to B - Make sure after the transaction the account information have to be updated for both the credit account and the debited account
- Also fetch the details of the account holder who are related from the same city
- Write a query to fetch account number and account holder name, whose accounts were created after 15th of any month
- Write a query to display the city name and count the branches in that city. Give the count of branches an alias name of Count_Branch.
- Write a query to display the account holder's id, account holder's name, branch id, and loan amount for people who have taken loans. (NOTE : use sql join concept to solve the query) Make sure to make your code clean kneat

Answer:

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/* =====
```

```
1. CREATE BANK TABLE
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===== */
```

```
CREATE TABLE Bank (
```

```
    branch_id INT PRIMARY KEY,
```

```
    branch_name VARCHAR(100) NOT NULL,
```

```
    branch_city VARCHAR(100) NOT NULL
```

);

/* =====

2. CREATE ACCOUNT_HOLDER TABLE

===== */

```
CREATE TABLE Account_Holder (  
    account_holder_id INT PRIMARY KEY,  
    account_no VARCHAR(20) UNIQUE NOT NULL,  
    account_holder_name VARCHAR(100) NOT NULL,  
    city VARCHAR(100) NOT NULL,  
    contact VARCHAR(15),  
    date_account_created DATE NOT NULL,  
    account_status VARCHAR(20) CHECK (account_status IN ('Active', 'Terminated')),  
    account_type VARCHAR(50),  
    balance DECIMAL(12,2) DEFAULT 0.00  
);
```

/* =====

3. CREATE LOAN TABLE

===== */

```
CREATE TABLE Loan (  
    loan_no INT PRIMARY KEY,  
    branch_id INT,  
    account_holder_id INT,
```

loan_amount DECIMAL(12,2),

loan_type VARCHAR(50),

FOREIGN KEY (branch_id) REFERENCES Bank(branch_id),

FOREIGN KEY (account_holder_id) REFERENCES Account_Holder(account_holder_id)

);

/* =====

4. INSERT SAMPLE DATA

===== */

INSERT INTO Bank VALUES (1, 'Main Branch', 'Mumbai');

INSERT INTO Bank VALUES (2, 'City Branch', 'Delhi');

INSERT INTO Bank VALUES (3, 'West Branch', 'Mumbai');

INSERT INTO Account_Holder VALUES

(101, 'ACC1001', 'Rahul Sharma', 'Mumbai', '9876543210', '2024-01-20', 'Active', 'Savings',
5000.00),

(102, 'ACC2002', 'Priya Verma', 'Mumbai', '9876501234', '2024-02-10', 'Active', 'Current',
3000.00),

(103, 'ACC3003', 'Amit Singh', 'Delhi', '9123456789', '2024-03-18', 'Active', 'Savings',
7000.00),

(104, 'ACC4004', 'Neha Kapoor', 'Delhi', '9988776655', '2024-04-25', 'Terminated', 'Savings',
2000.00);

INSERT INTO Loan VALUES

(1, 1, 101, 100000.00, 'Home Loan'),

(2, 2, 103, 50000.00, 'Car Loan');

/* =====

5. TRANSACTION - TRANSFER \$100 FROM ACC1001 TO ACC2002

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START TRANSACTION;

UPDATE Account_Holder

SET balance = balance - 100

WHERE account_no = 'ACC1001'

AND balance >= 100;

UPDATE Account_Holder

SET balance = balance + 100

WHERE account_no = 'ACC2002';

COMMIT;

/* =====

6. FETCH ACCOUNT HOLDERS FROM SAME CITY

===== */

SELECT *

FROM Account_Holder

```
WHERE city IN (  
    SELECT city  
    FROM Account_Holder  
    GROUP BY city  
    HAVING COUNT(*) > 1  
);
```

```
/* =====  
7. FETCH ACCOUNTS CREATED AFTER 15TH OF ANY MONTH  
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```

```
SELECT account_no, account_holder_name  
FROM Account_Holder  
WHERE DAY(date_account_created) > 15;
```

```
/* =====  
8. DISPLAY CITY NAME AND COUNT OF BRANCHES  
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```

```
SELECT branch_city, COUNT(branch_id) AS Count_Branch  
FROM Bank  
GROUP BY branch_city;
```

```
/* =====  
9. DISPLAY ACCOUNT HOLDER ID, NAME, BRANCH ID, LOAN AMOUNT
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(USING JOIN)

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SELECT

a.account_holder_id,

a.account_holder_name,

l.branch_id,

l.loan_amount

FROM Account_Holder a

JOIN Loan l

ON a.account_holder_id = l.account_holder_id;